**Log File:**

| Experiment Number | Parameters Chosen | Results |
|---|---|---|
| 1 | Neural Net:<br>Number of FC layers = 2 (24, 8)<br>Number of Activation layers = 2 (8, 1)<br>Neurons = 33<br>Error Function = RMSE<br>Regularization Parameter = NA<br>Activation Function = sigmoid<br>Epochs: 10<br>Learning Rate: 0.001 | Train/Test Split: 10:90<br>Size of dataset: 65188<br>Training Accuracy: 0.75<br>Test Accuracy: 0.75<br>Training RMSE: 0.258320<br>Test RMSE: 0.502321 |
| 2 | Neural Net:<br>Number of FC layers: 2 (24, 8)<br>Number of Activation layers: 2 (8, 1)<br>Neurons: 33<br>Error Function: RMSE<br>Activation Function: sigmoid<br>Epochs: 10<br>Learning Rate: 0.001 | Train/Test Split: 50:50<br>Size of dataset: 65188<br>Training Accuracy: 0.75<br>Test Accuracy: 0.74<br>Training RMSE: 0.250380<br>Test RMSE: 0.505841 |
| 3 | Neural Net:<br>Number of FC layers: 2 (24, 8)<br>Number of Activation layers: 2 (8, 1)<br>Neurons: 33<br>Error Function: RMSE<br>Regularization Parameter: NA<br>Activation Function: sigmoid<br>Epochs: 10<br>Learning Rate: 0.001 | Train/Test Split: 20:80<br>Size of dataset: 65188<br>Training Accuracy: 0.75<br>Test Accuracy: 0.75<br>Training RMSE: 0.253730<br>Test RMSE: 0.502892 |
| 4 | Neural Net:<br>Number of FC layers: 2 (24, 8)<br>Number of Activation layers: 2 (8, 1)<br>Neurons: 33<br>Error Function: RMSE<br>Regularization Parameter: NA<br>Activation Function: sigmoid<br>Epochs: 10<br>Learning Rate: 0.001 | Train/Test Split: 05/95<br>Size of dataset: 65188<br>Training Accuracy: 0.75<br>Test Accuracy: 0.75<br>Training RMSE: 0.270082<br>Test RMSE: 0.502235 |

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| 5 | Neural Net:<br>Number of FC layers: 2 (24, 8)(8, 1)<br>Number of Activation layers: 2<br>Neurons: 33<br>Error Function: RMSE<br>Regularization Parameter: NA<br>Activation Function: relu<br>Epochs: 10<br>Learning Rate: 0.001 | Train/Test Split: 05/95<br>Size of dataset: 65188<br>Training Accuracy: 0.75<br>Test Accuracy: 0.75<br>Training RMSE: 0.249463<br>Test RMSE: 0.502235 |
| 6 | Neural Net:<br>Number of FC layers: 4 (24, 50)(50, 100)(100, 25)(25, 1)<br>Number of Activation layers: 4<br>Neurons: 200<br>Error Function: RMSE<br>Regularization Parameter: NA<br>Activation Function: sigmoid, relu, sigmoid, sigmoid<br>Epochs: 10<br>Learning Rate: 0.005 | Train/Test Split: 05:95<br>Size of dataset: 65188<br>Training Accuracy: 0.75<br>Test Accuracy: 0.75<br>Training RMSE: 0.249183<br>Test RMSE: 0.502892 |
| AFTER RESAMPLING | | |
| 1 | Neural Net:<br>Number of FC layers: 4 (24, 250)(250, 200)(200, 50)(50, 1)<br>Number of Activation layers: 4 (sigmoid)<br>Neurons: 525<br>Error Function: RMSE<br>Epochs: 10<br>Learning Rate: 0.05 | Train/Test Split: 05:95<br>Size of dataset: 65188<br>Training Accuracy: 0.51<br>Test Accuracy: 0.50<br>Training RMSE: 0.490253<br>Test RMSE: 0.707469 |
| 2 | Neural Net:<br>Number of FC layers: 4 (24, 250)(250, 200)(200, 50)(50, 1)<br>Number of Activation layers: 4 (sigmoid)<br>Neurons: 525<br>Error Function: binary cross entropy<br>Epochs: 10<br>Learning Rate: 0.005 | Train/Test Split: 20:80<br>Size of dataset (neg/pos): 48754/48754<br>Training Accuracy: 0.51<br>Test Accuracy: 0.51<br>Training RMSE: 0.693005<br>Test RMSE: 0.698364 |
| 3 | Neural Net: | Train/Test Split: 20/80 |

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| | Number of FC layers: 3 (24, 12)(12, 8)(8, 1)<br>Number of Activation layers: 3<br>Neurons: 45<br>Error Function: binary cross entropy<br>Epochs: 15<br>Learning Rate: 0.01 | Size of dataset (neg/pos): 48754/48754<br>Training Accuracy: 0.56<br>Test Accuracy: 0.56<br>Training RMSE: 0.668180<br>Test RMSE: 0.662260 |
| 4 | Neural Net:<br>Number of FC layers: 3 (24, 12)(12, 8)(8, 1)<br>Number of Activation layers: 3<br>Neurons: 45<br>Error Function: binary cross entropy<br>Epochs: 30<br>Learning Rate: 0.01 | Train/Test Split: 20:80<br>Size of dataset (neg/pos): 16434/16434<br>Training Accuracy: 0.61<br>Test Accuracy: 0.60<br>Training RMSE: 0.638256<br>Test RMSE: 0.631353 |
| 5 | Neural Net:<br>Number of FC layers: 3 (24, 12)(12, 8)(8, 1)<br>Number of Activation layers: 3<br>Neurons: 45<br>Error Function: binary cross entropy<br>Epochs: 40<br>Learning Rate: 0.005 | Train/Test Split: 50:50<br>Size of dataset (neg/pos): 16434/25000<br>Training Accuracy: 0.62<br>Test Accuracy: 0.60<br>Training RMSE: 0.636644<br>Test RMSE: 0.630820 |
| 6 | Neural Net:<br>Number of FC layers: 3 (24, 12)(12, 8)(8, 1)<br>Number of Activation layers: 3<br>Neurons: 45<br>Error Function: binary cross entropy<br>Epochs: 40<br>Learning Rate: 0.005 | Train/Test Split: 80:20<br>Size of dataset (neg/pos): 16434/25000<br>Training Accuracy: 0.61<br>Test Accuracy: 0.61<br>Training RMSE: 0.624356<br>Test RMSE: 0.627206 |
| AFTER TWEAKING MODEL | | |
| 1 | Neural Net:<br>Number of FC layers: 4 (24, 250)(250, 200)(200, 50)(50, 1)<br>Number of Activation layers: 4 (sigmoid)<br>Neurons: 525<br>Error Function: RMSE<br>Epochs: 10 | Train/Test Split: 05:95<br>Size of dataset: 65188<br>Training Accuracy: 0.51<br>Test Accuracy: 0.50<br>Training RMSE: 0.490253<br>Test RMSE: 0.707469 |

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| | Learning Rate: 0.05 | |
| 2 | Neural Net:<br>Number of FC layers: 4 (24, 250)(250, 200)(200, 50)(50, 1)<br>Number of Activation layers: 4 (sigmoid)<br>Neurons: 525<br>Error Function: binary cross entropy<br>Epochs: 10<br>Learning Rate: 0.005 | Train/Test Split: 20:80<br>Size of dataset (neg/pos): 48754/48754<br>Training Accuracy: 0.51<br>Test Accuracy: 0.51<br>Training RMSE: 0.693005<br>Test RMSE: 0.698364 |
| 3 | Neural Net:<br>Number of FC layers: 3 (24, 12)(12, 8)(8, 1)<br>Number of Activation layers: 3<br>Neurons: 45<br>Error Function: binary cross entropy<br>Epochs: 15<br>Learning Rate: 0.01 | Train/Test Split: 20:80<br>Size of dataset (neg/pos): 16434/16434<br>Training Accuracy: 0.54<br>Test Accuracy: 0.54<br>Training RMSE: 0.668838<br>Test RMSE: 0.680117 |
| 4 | Neural Net:<br>Number of FC layers: 3 (24, 12)(12, 8)(8, 1)<br>Number of Activation layers: 3<br>Neurons: 45<br>Error Function: binary cross entropy<br>Epochs: 30<br>Learning Rate: 0.01 | Train/Test Split: 20:80<br>Size of dataset (neg/pos): 16434/16434<br>Training Accuracy: 0.59<br>Test Accuracy: 0.57<br>Training RMSE: 0.646651<br>Test RMSE: 0.659046 |
| 5 | Neural Net:<br>Number of FC layers: 3 (24, 12)(12, 8)(8, 1)<br>Number of Activation layers: 3<br>Neurons: 45<br>Error Function: binary cross entropy<br>Epochs: 40<br>Learning Rate: 0.005 | Train/Test Split: 50:50<br>Size of dataset (neg/pos): 16434/16434<br>Training Accuracy: 0.60<br>Test Accuracy: 0.58<br>Training RMSE: 0.649471<br>Test RMSE: 0.651246 |
| 6 | Neural Net:<br>Number of FC layers: 3 (24, 12)(12, 8)(8, 1)<br>Number of Activation layers: 3<br>Neurons: 45<br>Error Function: binary cross entropy<br>Epochs: 40<br>Learning Rate: 0.005 | Train/Test Split: 80:20<br>Size of dataset (neg/pos): 16434/16434<br>Training Accuracy: 0.61<br>Test Accuracy: 0.60<br>Training RMSE: 0.636222<br>Test RMSE: 0.632022 |